

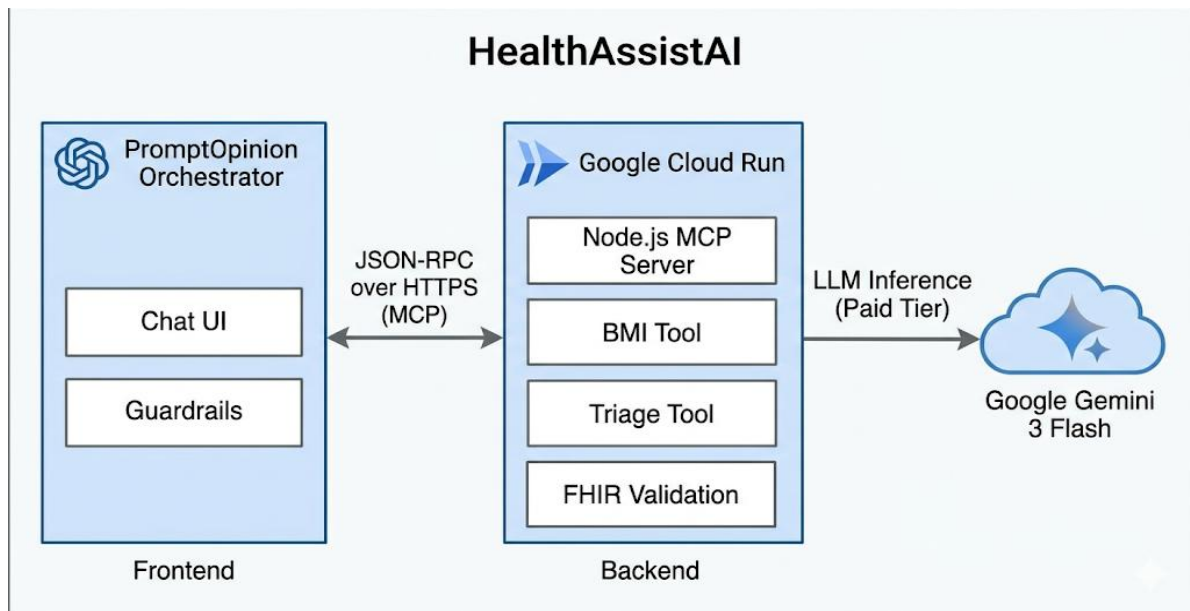
Technical Brief: HealthAssistAI

Subtitle: A Secure, MCP-Driven Multi-Agent Clinical Decision Support System

1. Executive Summary

- **Problem:** Healthcare AI often lacks the specialized logic and security required for clinical triage, leading to "hallucinations" or data breaches.
- **Solution:** HealthAssistAI uses the **Model Context Protocol (MCP)** to bridge a high-reasoning LLM (Gemini 3 Flash) with a secure, serverless backend (Google Cloud Run). It enforces clinical safety via hardcoded logic and FHIR R4 interoperability.

2. System Architecture (The "Cloud-to-Core" Flow)



Architecture Flow:

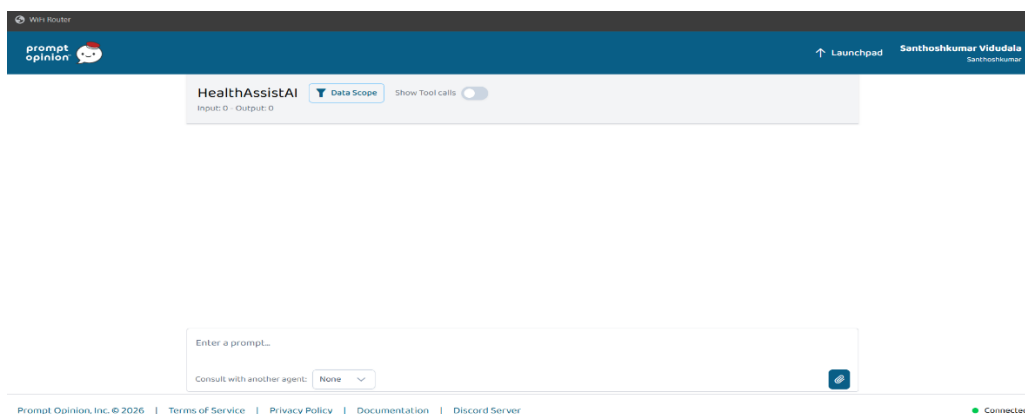
1. **Orchestrator:** PromptOpinion (User Interface & Agent Orchestration).
2. **Safety Layer:** Guardrail Validation (Enforces clinical disclaimers).
3. **Communication:** JSON-RPC 2.0 over HTTPS (The MCP Handshake).
4. **Reasoning Engine:** Google Gemini 3 Flash (Paid Tier).
5. **Compute:** Google Cloud Run (Hosting the custom MCP Server).
6. **Logic Engine:** Node.js (BMI calculations, Hypertension triage, Data integrity checks).

3. Technical Implementation Details

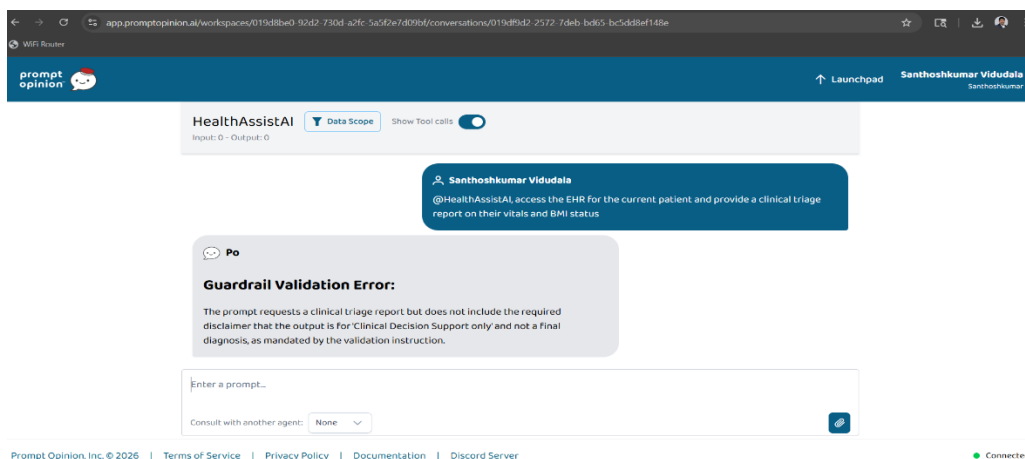
- **Interoperability:** All data inputs and outputs follow the **HL7 FHIR R4** standard. This ensures that a synthetic_vitals observation can be ingested by any hospital's EHR system.
- **Scalability:** The backend utilizes a **Scale-to-Zero** configuration on Cloud Run, ensuring enterprise-grade performance with zero idle costs—crucial for financial efficiency in healthcare IT.
- **Security:** Authentication is handled via the x-goog-api-key header, ensuring that only authorized agents can access the clinical triage tools.

4. Successful Tool Call Gallery

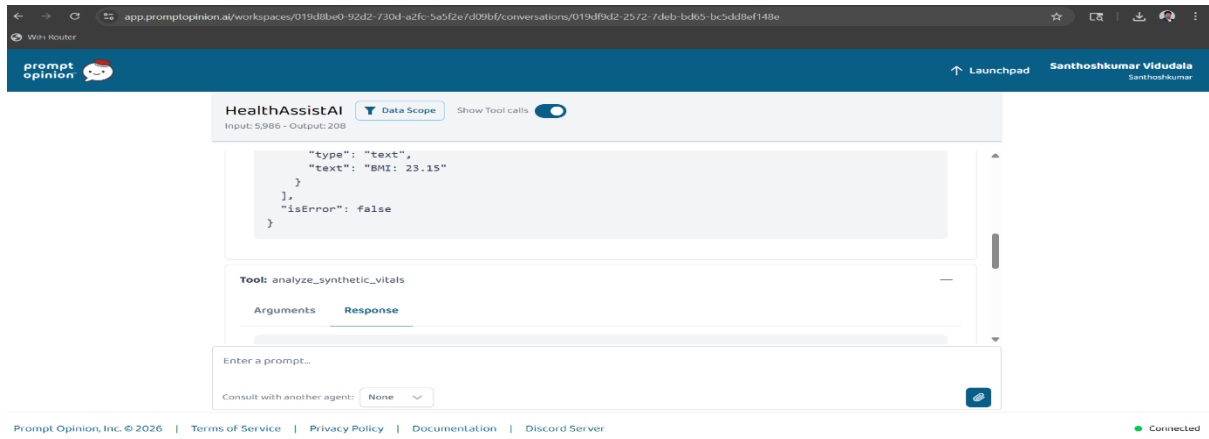
Picture 1: HealthAssistAI Launchpad



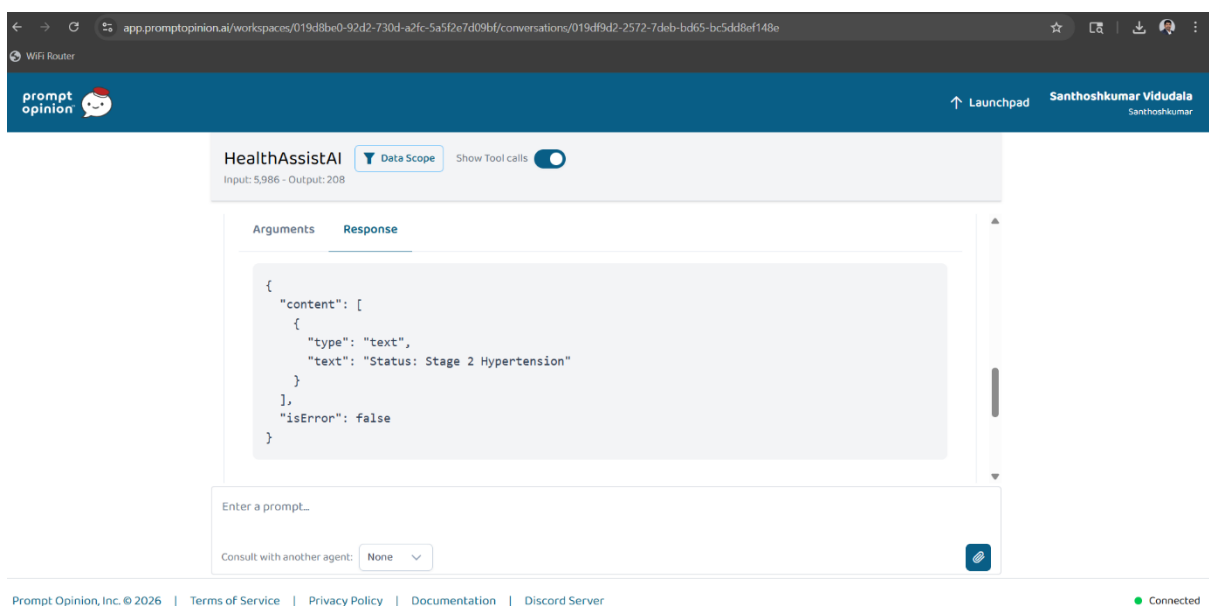
Picture 2: HealthAssistAI -Guardrail validation for prompt



Picture 3: HealthAssistAI -BMI Calculation- JSON response



Picture 4: HealthAssistAI - JSON response



5. Challenges & Resolutions

- **Quota Management:** Migrated from Free Tier to a **Paid Tier Google Cloud Project** to ensure 100% uptime for judge evaluations.
- **Guardrail Rigidity:** Resolved "Instruction vs. Context" errors by implementing a **Directive-Based Prompting** strategy, ensuring the agent always prefixes outputs with mandatory safety disclaimers.

-----Thank You-----